

ABSTRACT

A computer-implemented system and method for adaptive network configuration deployment based on physical state metrics is disclosed. The system collects device physical metrics including temperature, CPU load, and hardware aging indicators. A damping factor is calculated using the physical metrics and is used to dynamically adjust deployment concurrency and execution timing. The network topology is decomposed into subdomains, enabling parallel deployment while maintaining convergence stability. Convergence is monitored using a residual defined as the difference between actual device state and target configuration state. The system prevents device overload, improves deployment stability, and provides deterministic convergence behavior. The invention is applicable to data centers, telecom networks, cloud infrastructure, and distributed computing environments.